

Lecture 9 – Resampling Methods

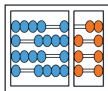
[ISLP 5.1–5.2]

MC886 – Machine Learning

Marcelo S. Reis [†]

[†] Departamento de Sistemas de Informação
Instituto de Computação, Universidade Estadual de Campinas

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Today's summary

The validation set approach

Leave-One-Out Cross Validation (LOOCV)

k -fold cross validation

The bootstrap

Resampling methods

- ❖ They consist in repeatedly drawing samples from the training set and, for each draw, using it to refit the model
- ❖ Very useful for model assessment and/or selection
- ❖ Two important approaches:
 - **Cross validation**
 - **Bootstrap**

The validation set approach

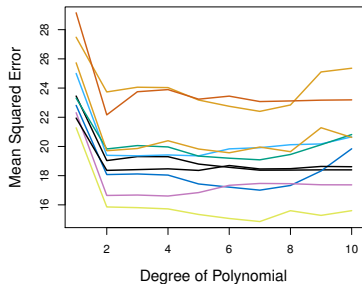
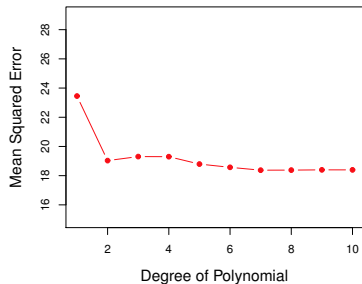
- ❖ We split training data into training and *validation* sets (Test data is still kept aside)
- ❖ We fit the model parameters using this smaller training set
- ❖ The validation set is used to estimate the out-of-sample error

Random split of n points into training and validation sets



Numbers represent indices of the i th observation (x_i, y_i)

Validation set approach on **Auto** data



Left: MSE in validation set as a function of polynomial degree of the fitted model

Right: The same procedure repeated for 10 different random data splits into training and validation sets

Limitations of the validation set approach

Although this approach is very simple, it has some shortcomings:

1. Validation set estimation of the test error can be highly variable (remember the right graph of the previous figure)
2. We can have an overestimation of the test error because part of the data is not used for training

Leave-One-Out Cross Validation (LOOCV)

1. We select (x_1, y_1) to be the validation set and use the remaining points $\{(x_2, y_2), \dots, (x_n, y_n)\}$ as the training set
2. Using the training set, we fit a model and compute $MSE_1 = (y_1 - \hat{y}_1)^2$
3. We repeat the previous step with (x_2, y_2) in the place of (x_1, y_1) , thus computing MSE_2
4. Repeating this approach for all observations, we can have an average of these n test error estimates:

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n MSE_i \quad (5.1)$$

Leave-One-Out Cross Validation (LOOCV)



Pros and cons of LOOCV

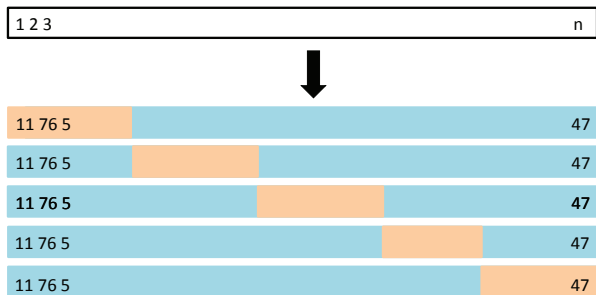
- 😊 $n - 1$ observations are used in training (far less bias than the validation set approach, which overestimates test error)
- 😊 LOOCV is deterministic (results don't rely on a random split into training and validation sets)
- 😞 Except for a few models (e.g., linear regression), there is no closed-form solution for LOOCV
 - We need to refit a model n times! (computationally expensive)

k -fold cross validation

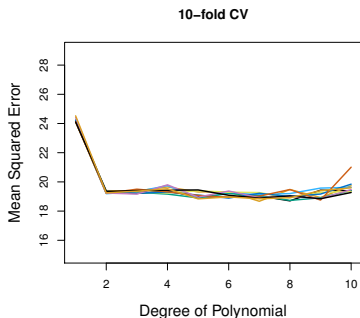
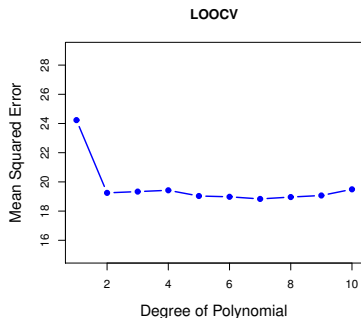
1. We randomly divide the data into k groups (folds)
2. We hold out fold 1, train with the remaining $k - 1$ folds and use fold 1 to compute the validation error MSE_1
3. We repeat the previous step with fold 2 in the place of fold 1, thus computing MSE_2
4. Repeating this approach for all folds, we can have an average of these k test error estimates:

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^k MSE_i \quad (5.3)$$

Example: 5-fold CV



CV on Auto data

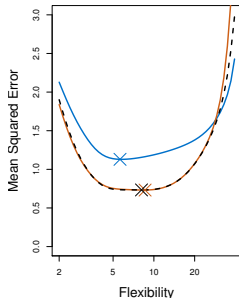


Left: LOOCV error curve

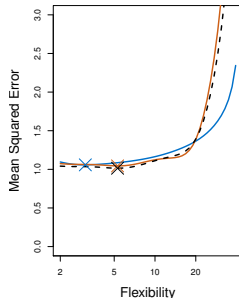
Right: 10-fold cross validation run for 9 different fold splits

True and estimated test MSE for simulated datasets

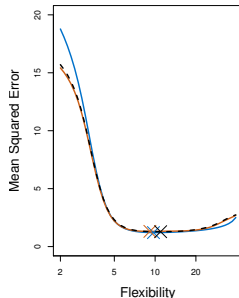
f approx. cubic



f approx. linear



f highly nonlinear



Blue line: true test MSE

Black dashed line: LOOCV test MSE estimate

Orange line: 10-fold test MSE estimate

X: minimum of each curve

Validation set vs. LOOCV vs. k -fold

- ❖ Validation set: high bias (overestimation of the test error)
- ❖ LOOCV: small bias; *however, it suffers of high variance (the mean of highly-correlated variables has high variance)*
- ❖ k -fold: a trade-off between validation set and LOOCV

CV on classification problems

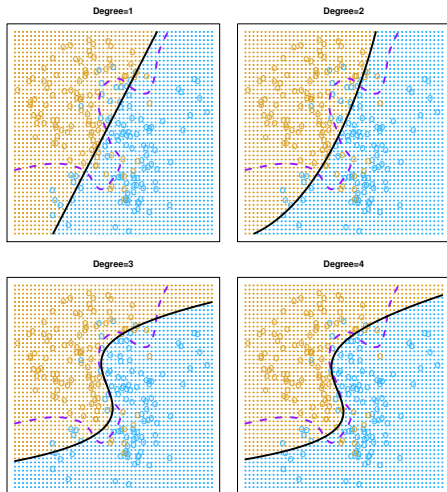
In classification settings, LOOCV takes the form of:

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n Err_i, \quad (5.4)$$

where $Err_i = I(y_i \neq \hat{y}_i)$

❖ k -fold is adapted in a similar way

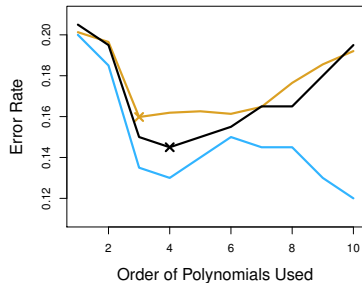
Polynomial logistic regression fit on simulated data ($p = 2$)



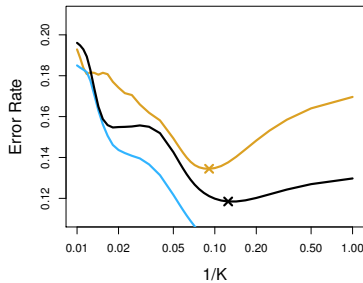
Purple dashed line: Bayes decision boundary

Using CV for choosing level of flexibility,
for the same simulated data of the previous slide

Polynomial logistic regression



KNN



Brown line: true test error
Black line: 10-fold CV error
Blue line: training error
X: minimum of the curve

The bootstrap

- ❖ A powerful statistical tool for quantifying the uncertainty associated with an estimator or model
 - e.g.: it can be used to estimate standard errors of the parameters of a linear regression
- ❖ Bootstrap can be used in a wide range of models (e.g., bagging, random forests)
- ❖ We'll study how it works with a toy model example

Toy model for bootstrap: investment allocation

- ❖ Suppose we want to invest a fixed sum of money in two financial assets that yield returns X and Y , respectively
- ❖ We'll invest a fraction α in X and $1 - \alpha$ in Y
- ❖ We are interested in minimizing the total risk (variance) of our investment, which means we want to minimize:

$$\text{Var}(\alpha X + (1 - \alpha)Y)$$

Minimization of the total risk (variance)

- ❖ The α that minimizes $\text{Var}(\alpha X + (1 - \alpha)Y)$ is:

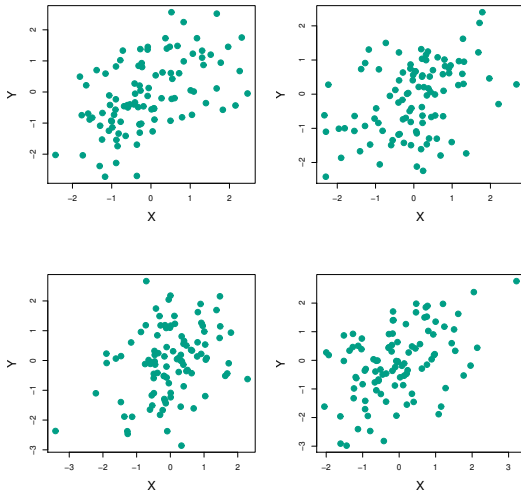
$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}, \quad (5.6)$$

where $\sigma_X^2 = \text{Var}(X)$, $\sigma_Y^2 = \text{Var}(Y)$ and $\sigma_{XY} = \text{Cov}(X, Y)$

- ❖ σ_X^2 , σ_Y^2 and σ_{XY} are unknown

(they must be estimated from data composed of past measurements of X and Y)

Each panel has 100 simulated returns for X and Y



They can be used to compute $\hat{\alpha}_1, \dots, \hat{\alpha}_4$

Computing the standard error (SE) for $\hat{\alpha}$

- ❖ We can use SE to quantify the accuracy for our estimate of α
- ❖ To this end, we can repeat the simulation procedure 1000 times, thus getting $\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_{1000}$
- ❖ Then the SE can be computed using:

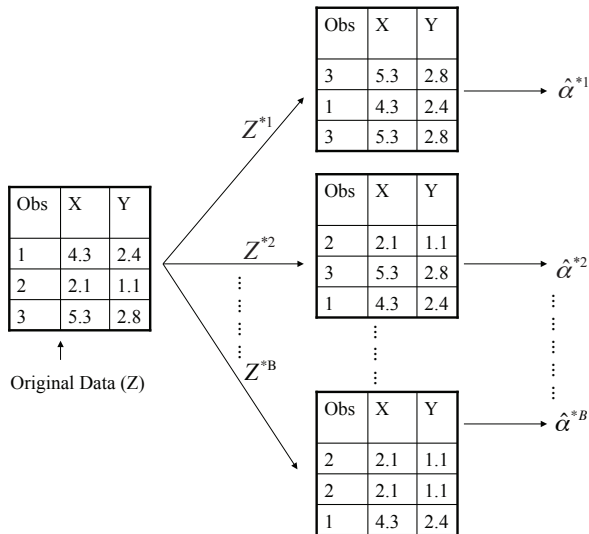
$$SE(\hat{\alpha}) = \sqrt{\frac{1}{1000 - 1} \sum_{r=1}^{1000} (\hat{\alpha}_r - \bar{\alpha})^2},$$

$$\text{where } \bar{\alpha} = \frac{1}{1000} \sum_{r=1}^{1000} \hat{\alpha}_r$$

However, we have a problem...

- ❖ ...in practical situations, we have a single dataset!
- ❖ How to deal with it?
- ❖ **Idea:** instead of getting more datasets, let's use our one for creating new datasets through *sampling with replacement!!!*
- ❖ **Underlying key principle:** the dataset has a sample distribution that resembles the population distribution
 - Hence, we can use the dataset to draw random samples in a similar way we would do using the population distribution!

Creation of B bootstrap datasets



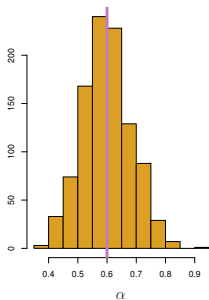
Computing the SE for $\hat{\alpha}$ with bootstrap

- ❖ Use bootstrap to compute B datasets $Z^{*1}, \dots, Z^{*B}$
- ❖ Those datasets are used to get $\hat{\alpha}^{*1}, \hat{\alpha}^{*2}, \dots, \hat{\alpha}^{*B}$
- ❖ Finally the SE can be computed using:

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1} \sum_{r=1}^B \left(\hat{\alpha}^r - \frac{1}{B} \sum_{r'=1}^B \hat{\alpha}^{*r'} \right)^2} \quad (5.8)$$

$\hat{\alpha}$ accuracy quantification by 1000 datasets vs. bootstrap

1000 datasets



B bootstrap datasets

